# Операционные системы

## Основы интерфейса взаимодействия пользователя с системой Unix на уровне командной строки

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# 1. Цели и задачи работы

# Цель лабораторной работы

Приобретение практических навыков взаимодействия пользователя с системой посредством командной строки.

# Задачи лабораторной работы

1 Определить имя и путь домашнего каталога

2 Изучить команду ls.

3 Выполнить действия с каталогами.

4 Получить дополнительные сведения при помощи справки по командам.

5 Изучить команду history.

# 2. Процесс выполнения лабораторной работы

# Имя и путь к домашнему каталогу

```
aatolpina@aatolpina:~$ cd
aatolpina@aatolpina:~$ pws
bash: pws: command not found...
aatolpina@aatolpina:~$ pwd
/home/aatolpina
aatolpina@aatolpina:~$
```

**Рисунок 1:** Путь к домашнему каталогу

# Опции команды ls

```
aatolpina@aatolpina:~$ cd /tmp
aatolpina@aatolpina:/tmp$ ls
a0c68a19-82d6-4f58-bb13-d9bdf22a60dd.zip
b92e84b6-98d9-4700-af42-66bb4cef7b0c.zip
happd.sock
snap-private-tmp
systemd-private-ebe93dd246864f95a035f7c846388961-abrtd.service-IWvLXe
systemd-private-ebe93dd246864f95a035f7c846388961-chronyd.service-tLkduX
systemd-private-ebe93dd246864f95a035f7c846388961-colord.service-EZuOe0
systemd-private-ebe93dd246864f95a035f7c846388961-dbus-broker.service-rJEzu1
systemd-private-ebe93dd246864f95a035f7c846388961-fwupd.service-mppSg4
systemd-private-ebe93dd246864f95a035f7c846388961-irqbalance.service-R7j48Q
systemd-private-ebe93dd246864f95a035f7c846388961-low-memory-monitor.service-WhS8SC
systemd-private-ebe93dd246864f95a035f7c846388961-ModemManager.service-Xnej8b
systemd-private-ebe93dd246864f95a035f7c846388961-pcscd.service-96Ttt0
systemd-private-ebe93dd246864f95a035f7c846388961-polkit.service-HRmxyy
systemd-private-ebe93dd246864f95a035f7c846388961-rtkit-daemon.service-HkjYSm
systemd-private-ebe93dd246864f95a035f7c846388961-switcheroo-control.service-yFzqWj
systemd-private-ebe93dd246864f95a035f7c846388961-systemd-logind.service-ovmVwU
systemd-private-ebe93dd246864f95a035f7c846388961-systemd-resolved.service-ZKCOKW
systemd-private-ebe93dd246864f95a035f7c846388961-upower.service-LjvZFP
tmp-d05.xpi
vmware-root_904-2697008433
aatolpina@aatolpina:/tmp$
```

**Рисунок 2:** Команда ls

# Опции команды ls

```
aatolpina@aatolpina:/tmp$ ls -a
.
..
a0c68a19-82d6-4f58-bb13-d9bdf22a60dd.zip
b92e84b6-98d9-4700-af42-66bb4cef7b0c.zip
.font-unix
happd.sock
.ICE-unix
snap-private-tmp
systemd-private-ebe93dd246864f95a035f7c846388961-abrtd.service-IWvLXe
systemd-private-ebe93dd246864f95a035f7c846388961-chronyd.service-tLkduX
systemd-private-ebe93dd246864f95a035f7c846388961-colord.service-EZuOe0
systemd-private-ebe93dd246864f95a035f7c846388961-dbus-broker.service-rJEzu1
systemd-private-ebe93dd246864f95a035f7c846388961-fwupd.service-mppSg4
systemd-private-ebe93dd246864f95a035f7c846388961-irqbalance.service-R7j48Q
systemd-private-ebe93dd246864f95a035f7c846388961-low-memory-monitor.service-WhS8SC
systemd-private-ebe93dd246864f95a035f7c846388961-ModemManager.service-Xnej8b
systemd-private-ebe93dd246864f95a035f7c846388961-pcscd.service-9GTtt0
systemd-private-ebe93dd246864f95a035f7c846388961-polkit.service-HRmxyy
systemd-private-ebe93dd246864f95a035f7c846388961-rtkit-daemon.service-HkjYSm
systemd-private-ebe93dd246864f95a035f7c846388961-switcheroo-control.service-yFzqWj
systemd-private-ebe93dd246864f95a035f7c846388961-systemd-logind.service-ovmVwU
systemd-private-ebe93dd246864f95a035f7c846388961-systemd-resolved.service-ZKCOKW
systemd-private-ebe93dd246864f95a035f7c846388961-upower.service-LjvZFP
tmp-d05.xpi
vmware-root_904-2697008433
.X0-lock
.X11-unix
.X1-lock
.XIM-unix
aatolpina@aatolpina:/tmp$
```

**Рисунок 3:** Команда ls -a

# Опции команды ls

```
aatolpina@aatolpina:/tmp$ ls -l
total 9696
-rw-r--r--. 1 aatolpina aatolpina 4182728 Mar  9 17:31 a0c68a19-82d6-4f58-bb13-d9bdf22a60dd.zip
-rw-r--r--. 1 aatolpina aatolpina 4182728 Mar  9 17:31 b92e84b6-98d9-4700-af42-66bb4cef7b0c.zip
srw-rw-rw-. 1 root      root            0 Mar  9 17:15 happd.sock
drwx------. 2 root      root           40 Mar  9 17:14 snap-private-tmp
drwx------. 3 root      root           60 Mar  9 17:14 systemd-private-ebe93dd246864f95a035f7c84638
8961-abrtd.service-IWvLXe
drwx------. 3 root      root           60 Mar  9 17:14 systemd-private-ebe93dd246864f95a035f7c84638
8961-chronyd.service-tLkduX
drwx------. 3 root      root           60 Mar  9 17:15 systemd-private-ebe93dd246864f95a035f7c84638
8961-colord.service-EZuOe0
drwx------. 3 root      root           60 Mar  9 17:14 systemd-private-ebe93dd246864f95a035f7c84638
8961-dbus-broker.service-rJEzu1
drwx------. 3 root      root           60 Mar  9 17:30 systemd-private-ebe93dd246864f95a035f7c84638
8961-fwupd.service-mppSg4
drwx------. 3 root      root           60 Mar  9 17:14 systemd-private-ebe93dd246864f95a035f7c84638
8961-irqbalance.service-R7j48Q
drwx------. 3 root      root           60 Mar  9 17:14 systemd-private-ebe93dd246864f95a035f7c84638
8961-low-memory-monitor.service-WhS8SC
drwx------. 3 root      root           60 Mar  9 17:14 systemd-private-ebe93dd246864f95a035f7c84638
8961-ModemManager.service-Xnej8b
drwx------. 3 root      root           60 Mar  9 17:15 systemd-private-ebe93dd246864f95a035f7c84638
8961-pcscd.service-96Ttt0
drwx------. 3 root      root           60 Mar  9 17:14 systemd-private-ebe93dd246864f95a035f7c84638
8961-polkit.service-HRmxyy
drwx------. 3 root      root           60 Mar  9 17:14 systemd-private-ebe93dd246864f95a035f7c84638
8961-rtkit-daemon.service-HkiYSm
```

**Рисунок 4:** Команда ls -l

# Опции команды ls

```
aatolpina@aatolpina:/tmp$ ls -f
.
..
a0c68a19-82d6-4f58-bb13-d9bdf22a60dd.zip
tmp-d05.xpi
b92e84b6-98d9-4700-af42-66bb4cef7b0c.zip
systemd-private-ebe93dd246864f95a035f7c846388961-fwupd.service-mppSg4
.X1-lock
.X0-lock
systemd-private-ebe93dd246864f95a035f7c846388961-pcscd.service-96Ttt0
systemd-private-ebe93dd246864f95a035f7c846388961-colord.service-EZuOe0
happd.sock
systemd-private-ebe93dd246864f95a035f7c846388961-ModemManager.service-Xnej8b
vmware-root_904-2697008433
systemd-private-ebe93dd246864f95a035f7c846388961-systemd-logind.service-ovmVwU
systemd-private-ebe93dd246864f95a035f7c846388961-upower.service-LjvZFP
systemd-private-ebe93dd246864f95a035f7c846388961-switcheroo-control.service-yFzqWj
systemd-private-ebe93dd246864f95a035f7c846388961-rtkit-daemon.service-HkjYSm
systemd-private-ebe93dd246864f95a035f7c846388961-polkit.service-HRmxyy
systemd-private-ebe93dd246864f95a035f7c846388961-low-memory-monitor.service-WhS8SC
systemd-private-ebe93dd246864f95a035f7c846388961-chronyd.service-tLkduX
systemd-private-ebe93dd246864f95a035f7c846388961-irqbalance.service-R7j48Q
systemd-private-ebe93dd246864f95a035f7c846388961-abrtd.service-IWvLXe
systemd-private-ebe93dd246864f95a035f7c846388961-dbus-broker.service-rJEzu1
systemd-private-ebe93dd246864f95a035f7c846388961-systemd-resolved.service-ZKCOKW
.font-unix
.XIM-unix
.ICE-unix
.X11-unix
snap-private-tmp
aatolpina@aatolpina:/tmp$
```

**Рисунок 5:** Команда ls -f

# Каталог /var/spool

```
aatolpina@aatolpina:/tmp$ cd /var/spool/
aatolpina@aatolpina:/var/spool$ ls -l
total 0
drwxr-x--x. 1 root abrt 10200 Mar  9 17:31 abrt
drwx------. 1 abrt abrt     0 Dec  4 03:00 abrt-upload
drwxr-xr-x. 1 root root    66 Oct 23 07:05 anacron
drwx------. 1 root root    18 Oct 23 07:07 at
drwx------. 1 root root     0 Aug  6  2025 cron
drwx--x---. 1 root lp       6 Dec 12 03:00 cups
drwxr-xr-x. 1 root root     0 Jul 30  2025 lpd
drwxrwxr-x. 1 root mail  1466 Mar  7 12:48 mail
drwxr-xr-x. 1 root root     0 Jan 27 03:00 plymouth
aatolpina@aatolpina:/var/spool$
```

**Рисунок 6:** Каталог /var/spool

# Домашний каталог

```
aatolpina@aatolpina:/var/spool$ cd
aatolpina@aatolpina:~$ ls
Desktop    Downloads     Music     Public  snap       Videos
Documents  git-extended  Pictures  site    Templates  work
aatolpina@aatolpina:~$ ls -al
total 24
drwx------. 1 aatolpina aatolpina  382 Feb 27 16:21 .
drwxr-xr-x. 1 root      root      1466 Mar  7 12:48 ..
-rw-------. 1 aatolpina aatolpina 5118 Feb 27 16:34 .bash_history
-rw-r--r--. 1 aatolpina aatolpina   18 Jul 23  2025 .bash_logout
-rw-r--r--. 1 aatolpina aatolpina  144 Jul 23  2025 .bash_profile
-rw-r--r--. 1 aatolpina aatolpina  679 Feb 27 16:11 .bashrc
drwx------. 1 aatolpina aatolpina  492 Feb 27 16:17 .cache
drwx------. 1 aatolpina aatolpina  364 Feb 27 16:34 .config
drwxr-xr-x. 1 aatolpina aatolpina    0 Feb 27 15:58 Desktop
drwxr-xr-x. 1 aatolpina aatolpina    0 Feb 27 15:58 Documents
drwxr-xr-x. 1 aatolpina aatolpina    0 Feb 27 15:58 Downloads
-rw-r--r--. 1 aatolpina aatolpina  232 Feb 27 16:06 .gitconfig
drwxr-xr-x. 1 aatolpina aatolpina   74 Feb 27 16:14 git-extended
drwx------. 1 aatolpina aatolpina  136 Feb 27 16:05 .gnupg
drwx------. 1 aatolpina aatolpina   20 Feb 27 15:58 .local
drwxr-xr-x. 1 aatolpina aatolpina   34 Oct 23 07:05 .mozilla
drwxr-xr-x. 1 aatolpina aatolpina    0 Feb 27 15:58 Music
drwxr-xr-x. 1 aatolpina aatolpina   26 Feb 27 16:21 .npm
drwxr-xr-x. 1 aatolpina aatolpina    0 Feb 27 15:58 Pictures
drwxr-xr-x. 1 aatolpina aatolpina    0 Feb 27 15:58 Public
```

**Рисунок 7:** Файлы в домашнем каталоге

# Работа с каталогами

```
aatolpina@aatolpina:~$ mkdir newdir
aatolpina@aatolpina:~$ mkdir newdir/morefun
aatolpina@aatolpina:~$ mkdir letters memos misk
aatolpina@aatolpina:~$ ls
Desktop    Downloads     letters  misk   newdir    Public  snap       Videos
Documents  git-extended  memos    Music  Pictures  site    Templates  work
aatolpina@aatolpina:~$ rm letters/ memos/ misk/
rm: cannot remove 'letters/': Is a directory
rm: cannot remove 'memos/': Is a directory
rm: cannot remove 'misk/': Is a directory
aatolpina@aatolpina:~$ rm -r letters/ memos/ misk/
aatolpina@aatolpina:~$ rm -r newdir/
aatolpina@aatolpina:~$ ls
Desktop    Downloads     Music    Public  snap       Videos
Documents  git-extended  Pictures site    Templates  work
aatolpina@aatolpina:~$
```

**Рисунок 8:** Действия с каталогами

# Опции команды ls

```
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_extens
ions/yamadharma/minted-quarto':
_extension.yml  minted-quarto.lua

'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/image':
solvay.jpg

'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resour
ces':
csl  tex

'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resour
ces/csl':
gost-r-7-0-5-2008-numeric.csl

'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resour
ces/tex':
preamble.tex

'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/scripts':
image-report  mpv-shot
aatolpina@aatolpina:~$ ls -t
snap  git-extended  Desktop    Downloads  Pictures  Templates
site  work          Documents  Music      Public    Videos
aatolpina@aatolpina:~$
```

**Рисунок 9:** Команда ls -R и ls -t

# Справка по командам

```
BASH_BUILTINS(1)                    General Commands Manual                    BASH_BUILTINS(1)

NAME
       :, ., [, alias, bg, bind, break, builtin, caller, cd, command, compgen, complete, compopt,
       continue,  declare,  dirs,  disown, echo, enable, eval, exec, exit, export, false, fc, fg,
       getopts, hash, help, history, jobs, kill,  let,  local,  logout,  mapfile,  popd,  printf,
       pushd,  pwd,  read, readarray, readonly, return, set, shift, shopt, source, suspend, test,
       times, trap, true, type, typeset, ulimit, umask, unalias, unset, wait - bash built-in com-
       mands, see bash(1)

BASH BUILTIN COMMANDS
       Unless otherwise noted, each builtin command documented in this section as  accepting  op-
       tions preceded by - accepts -- to signify the end of the options.  The :, true, false, and
       test/[  builtins  do  not accept options and do not treat -- specially.  The exit, logout,
       return, break, continue, let, and shift builtins accept and  process  arguments  beginning
       with  -  without requiring --.  Other builtins that accept arguments but are not specified
       as accepting options interpret arguments beginning with - as invalid options  and  require
       -- to prevent this interpretation.

       : [arguments]
              No  effect;  the command does nothing beyond expanding arguments and performing any
              specified redirections.  The return status is zero.

       . [-p path] filename [arguments]
       source [-p path] filename [arguments]
              The . command (source) reads and execute commands  from  filename  in  the  current
              shell  environment  and  returns  the exit status of the last command executed from
              filename.
 Manual page cd(1) line 1 (press h for help or q to quit)
```

**Рисунок 10:** Справка по команде cd

# Справка по командам

```
PWD(1)                                User Commands                                PWD(1)

NAME
       pwd - print name of current/working directory

SYNOPSIS
       pwd [OPTION]...

DESCRIPTION
       Print the full filename of the current working directory.

       -L, --logical
              use PWD from environment, even if it contains symlinks

       -P, --physical
              resolve all symlinks

       --help display this help and exit

       --version
              output version information and exit

       If no option is specified, -P is assumed.

       Your shell may have its own version of pwd, which usually supersedes the version described
       here.   Please  refer  to your shell's documentation for details about the options it sup-
       ports.

 Manual page pwd(1) line 1 (press h for help or q to quit)
```

**Рисунок 11:** Справка по команде pwd

# Справка по командам

```
MKDIR(1)                          User Commands                          MKDIR(1)

NAME
       mkdir - make directories

SYNOPSIS
       mkdir [OPTION]... DIRECTORY...

DESCRIPTION
       Create the DIRECTORY(ies), if they do not already exist.

       Mandatory arguments to long options are mandatory for short options too.

       -m, --mode=MODE
              set file mode (as in chmod), not a=rwx - umask

       -p, --parents
              no  error if existing, make parent directories as needed, with their file modes un-
              affected by any -m option

       -v, --verbose
              print a message for each created directory

       -Z     set SELinux security context of each created directory to the default type

       --context[=CTX]
              like -Z, or if CTX is specified then set the SELinux or SMACK security  context  to
              CTX
 Manual page mkdir(1) line 1 (press h for help or q to quit)
```

**Рисунок 12:** Справка по команде mkdir

# Справка по командам

```
RMDIR(1)                          User Commands                          RMDIR(1)

NAME
       rmdir - remove empty directories

SYNOPSIS
       rmdir [OPTION]... DIRECTORY...

DESCRIPTION
       Remove the DIRECTORY(ies), if they are empty.

       --ignore-fail-on-non-empty
              ignore each failure to remove a non-empty directory

       -p, --parents
              remove  DIRECTORY  and its ancestors; e.g., 'rmdir -p a/b' is similar to 'rmdir a/b
              a'

       -v, --verbose
              output a diagnostic for every directory processed

       --help display this help and exit

       --version
              output version information and exit

AUTHOR
       Written by David MacKenzie.
 Manual page rmdir(1) line 1 (press h for help or q to quit)
```

**Рисунок 13:** Справка по команде rmdir

# Справка по командам

```
RM(1)                                User Commands                                RM(1)

NAME
       rm - remove files or directories

SYNOPSIS
       rm [OPTION]... [FILE]...

DESCRIPTION
       This manual page documents the GNU version of rm.  rm removes each specified file.  By de-
       fault, it does not remove directories.

       If  the  -I  or --interactive=once option is given, and there are more than three files or
       the -r, -R, or --recursive are given, then rm prompts the user for whether to proceed with
       the entire operation.  If the response is not affirmative, the entire command is aborted.

       Otherwise, if a file is unwritable, standard input is a terminal, and the  -f  or  --force
       option  is  not  given,  or the -i or --interactive=always option is given, rm prompts the
       user for whether to remove the file.  If the response is  not  affirmative,  the  file  is
       skipped.

OPTIONS
       Remove (unlink) the FILE(s).

       -f, --force
              ignore nonexistent files and arguments, never prompt

       -i     prompt before every removal
 Manual page rm(1) line 1 (press h for help or q to quit)
```

**Рисунок 14:** Справка по команде rm

# История командной строки

```
276  cd /tmp
277  ls
278  ls -a
279  ls -l
280  ls -f
281  cd /var/spool/
282  ls -l
283  cd
284  ls
285  ls -al
286  mkdir newdir
287  mkdir newdir/morefun
288  mkdir letters memos misk
289  ls
290  rm letters/ memos/ misk/
291  rm -r letters/ memos/ misk/
292  rm -r newdir/
293  ls
294  ls -R
295  ls -t
296  help cd
297  man pwd
298  man mkdir
```

**Рисунок 15:** Команда history

# 3. Выводы по проделанной работе

# Вывод

Мы приобрели практические навыки взаимодействия пользователя с системой посредством командной строки.